# ONE NATION – ONE ELECTION

SMART VOTING • LIVE MONITORING • LEARNING THROUGH TECHNOLOGY

## PROJECT OVERVIEW

An educational polling simulator built with three ESP32 controllers: two Electronic Voting Machines (EVMs) representing **Dharapuram** and **Madurantakam**, and one **Central Dashboard** that collects their vote counts over a local Wi-Fi network and displays constituency-wise and combined results in a web browser.

| EVM 1<br>DHARAPURAM | EVM 2<br>MADURANTAKAM | CENTRAL DASHBOARD |
|---|---|---|
| Three party buttons<br>Stores vote counts<br>Serves live data over Wi-Fi | Three party buttons<br>Stores vote counts<br>Serves live data over Wi-Fi | Reads both EVM APIs<br>Combines totals<br>Displays live results |

## HOW IT WORKS

| 1. CAST A VOTE | 2. SEND DATA | 3. COMBINE & DISPLAY |
|---|---|---|
| Press DMK, AIADMK or TVK on either EVM. | Each EVM provides vote counts through a local web API. | The central ESP32 polls both stations and refreshes the dashboard. |

## KEY FEATURES

- Three voting options: DMK, AIADMK and TVK
- Vote counts saved in ESP32 non-volatile preferences
- Separate dashboards for each EVM
- Combined party totals and percentage bars
- Online / stale / waiting status for each EVM
- Automatic dashboard refresh approximately every 2 seconds

## TECHNOLOGY USED

**Hardware:** ESP32 development boards, push buttons, LED and buzzer. **Software:** Arduino IDE, C++, ESP32 Wi-Fi/WebServer libraries, ArduinoJson and browser-based HTML/CSS/JavaScript. **Connectivity:** Local Wi-Fi network (2.4 GHz Wi-Fi required by ESP32).

## LEARNING OUTCOMES

Embedded programming • IoT communication • Web dashboard design • JSON APIs • Data persistence • Real-time status monitoring • System integration • Problem-solving and responsible technology use.

**IMPORTANT NOTE**

This is an educational polling simulator created for demonstration and learning. It is **not** a certified electronic voting machine or an official election system. It should be used only on a trusted local network.

**DEMOCRACY · TECHNOLOGY · LEARNING FOR A BETTER TOMORROW**

**Developed by STREAM TEAM**